

Freeway Ramp Metering: An Overview

Markos Papageorgiou, *Fellow, IEEE*, and Apostolos Kotsialos

Abstract—Recurrent and nonrecurrent congestion on freeways may be alleviated if today's "spontaneous" infrastructure utilization is replaced by an orderly, controllable operation via comprehensive application of ramp metering and freeway-to-freeway control, combined with powerful optimal control techniques. This paper first explains why ramp metering can lead to a dramatic amelioration of traffic conditions on freeways. An overview of ramp metering algorithms is provided next, ranging from early fixed-time approaches to traffic-responsive regulators and to modern sophisticated nonlinear optimal control schemes. Finally, a large-scale example demonstrates the high potential of advanced ramp metering approaches.

Index Terms—Congestion management, nonlinear optimal control, ramp metering, traffic-responsive regulators.

I. INTRODUCTION

URBAN and interurban freeways had been originally conceived so as to provide virtually unlimited mobility to road users. The on-going dramatic expansion of car-ownership, however, has led to the daily appearance of recurrent and nonrecurrent freeway congestions of thousands of kilometers in length around the world. Ironically, daily recurrent congestions reduce substantially the available infrastructure capacity at the rush hours, i.e., at the time this capacity is most urgently needed, causing delays, increased environmental pollution, and reduced traffic safety. Similar effects are observed in the frequent case of nonrecurrent congestions caused by incidents, road works, etc. It has been recently realized that the mere infrastructure expansion cannot provide a complete solution to these problems due to economic and environmental reasons or, in metropolitan areas, simply due to lack of space.

The traffic situation on today's freeways resembles very much the one in urban road networks prior to the introduction of traffic lights: blocked links, chaotic intersections, reduced safety. It seems like road authorities and road users are still chasing the phantom of unlimited mobility that freeways were originally supposed to provide. What is urgently needed, however, is to restore and maintain the full utilization of the freeways' capacity along with an orderly and balanced satisfaction of the occurring demand both during rush hours and in case of incidents. Clearly, the passage from chaotic to optimal traffic conditions is only possible if today's "spontaneous" use of the freeway infrastructure is replaced by suitable control actions



Fig. 1. A general traffic network.

aiming at the benefit of all users. Ramp metering is the most efficient means to this end, whereby short delays at on-ramps and freeway-to-freeway intersections is the (relatively low) price to pay for capacity flow on the freeway itself, leading to substantial savings for each individual road user.

This overview paper first explains, based on simple, mathematically sound arguments, the reasons why ramp metering may lead to a substantial amelioration of traffic conditions on freeways (Section II). An overview of ramp metering algorithms is provided in Section III, ranging from early fixed-time approaches to traffic-responsive regulators and to modern, sophisticated nonlinear optimal control schemes. A simulation example of large-scale nonlinear optimal ramp metering is presented in Section IV to demonstrate the high amelioration potential of advanced ramp metering algorithms. Section V summarizes the main conclusions.

II. WHY RAMP METERING

A. A Basic Property

To be able to answer this question, we will first recall a simple fact. Consider any traffic network (Fig. 1) with demand appearing at several locations (e.g., at the on-ramps, in case of a freeway network) and exit flows forming at several destinations (e.g., at the freeway off-ramps). Clearly, the accumulated demand over, say, a day will be equal to the accumulated exit flows, because no vehicle disappears or is generated in the network. Let us assume that the demand level and its spatial and temporal distribution are independent of any control measures taken in the network. Then, we are interested to know how much accumulated time will be needed by all drivers to reach their respective destinations at the network exits (network efficiency!). It is quite evident that this **total time spent** by all drivers in the traffic network will be longer if, for any reason (e.g., due to lack of suitable control measures), the exit flows are temporarily lower, i.e. if vehicles are delayed within the network on their way to their destinations. As a consequence, any control measure or control strategy that can manage to increase the early exit flows of the network, will lead to a corresponding decrease of the total time spent.

The above statements may be formalized by use of simple mathematics [1], [2]. For the needs of this paper we will use a discrete-time representation of traffic variables with discrete

Manuscript received July 28, 2000; revised November 11, 2002. This work was supported in part by the European Commission in the framework of the project DACCORD (TR1017), Telematics Application Programme. The content of this paper is under the sole responsibility of the authors and in no way represents the views of the European Commission. The Associate Editor for this paper was P. A. Ioannou.

The authors are with Dynamic Systems and Simulation Laboratory, Technical University of Crete 73100 Chania, Greece (e-mail: markos@dssl.tuc.gr).

Digital Object Identifier 10.1109/TITS.2002.806803

time index $k = 0, 1, 2, \dots$ and time interval T . A **traffic volume** or **flow** $q(k)$ (in veh/h) is defined as the number of vehicles crossing a corresponding location during the time period $[kT, (k+1)T]$, divided by T .

We consider a traffic network (Fig. 1) that receives demands $d_i(k)$ (in veh/h) at its origins $i = 1, 2, \dots$ and we define the total demand $d(k) = d_1(k) + d_2(k) + \dots$. We assume that $d(k)$, $k = 0, \dots, K-1$, is independent of any control measures taken in the network. We define exit flows $s_i(k)$ at the network destinations $i = 1, 2, \dots$, and the total exit flow $s(k) = s_1(k) + s_2(k) + \dots$. We wish to apply control measures so as to minimize the total time spent T_s in the network over a time horizon K , i.e.

$$T_s = T \sum_{k=1}^K N(k) \quad (1)$$

where $N(k)$ is the total number of vehicles in the network at time k . Due to conservation of vehicles

$$\begin{aligned} N(k) &= N(k-1) + T[d(k-1) - s(k-1)] \\ &= N(0) + T \sum_{\kappa=0}^{k-1} [d(\kappa) - s(\kappa)]. \end{aligned} \quad (2)$$

Substituting (2) in (1) we obtain

$$T_s = T \sum_{k=1}^K \left[N(0) + T \sum_{\kappa=0}^{k-1} d(\kappa) - T \sum_{\kappa=0}^{k-1} s(\kappa) \right]. \quad (3)$$

The first two terms in the outer sum of (3) are independent of the control measures taken in the network, hence minimization of T_s is equivalent to maximization of the following quantity

$$S = T^2 \sum_{k=1}^K \sum_{\kappa=0}^{k-1} s(\kappa) = T^2 \sum_{\kappa=0}^{K-1} (K - \kappa) s(\kappa). \quad (4)$$

Thus minimization of the total time spent in a traffic network is equivalent to maximization of the time-weighted exit flows. In other words, the earlier the vehicles are able to exit the network (by appropriate use of the available control measures) the less time they will have spent in the network.

B. First Answer

We consider (see Fig. 2) two cases for a freeway on-ramp: (a) without and (b) with metering control. Let q_{in} be the upstream freeway flow, d be the ramp demand, q_{con} be the mainstream outflow in presence of congestion, and q_{cap} be the freeway capacity. It is well known that the outflow q_{con} in case of congestion is lower by some 5%–10% than the freeway capacity q_{cap} . In Fig. 2(b), we assume that ramp metering may be used to maintain capacity flow on the mainstream, e.g., by using the control strategy ALINEA [3] (see Section III-B). Of course, the application of ramp metering creates a queue at the on-ramp but, because q_{cap} is greater than q_{con} (increased outflow!), ramp metering leads to a reduction of the total time spent (including the ramp waiting time). It is easy to show [2] that the amelioration ΔT_s (in%) of the total time spent is given by

$$\Delta T_s = \frac{q_{cap} - q_{con}}{q_{in} + d - q_{con}} 100. \quad (5)$$

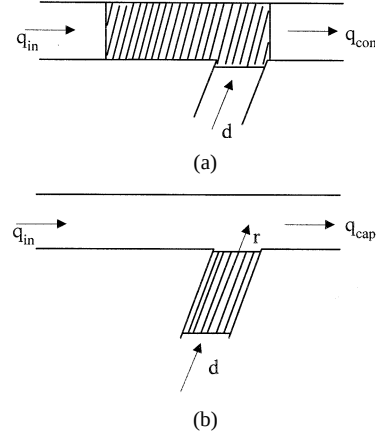


Fig. 2. Two cases: (a) without and (b) with ramp metering; grey areas indicate congestion zones.

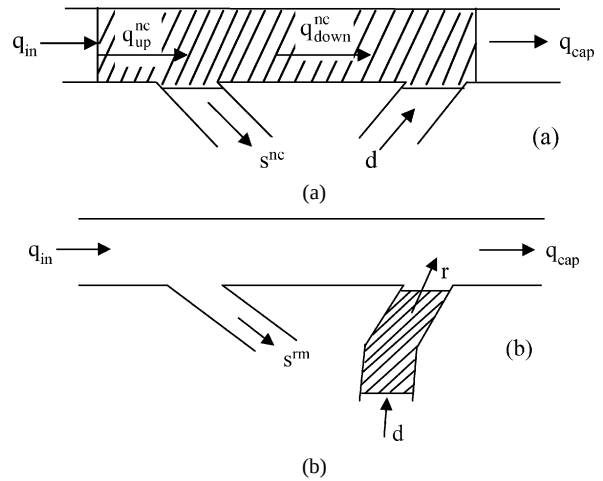


Fig. 3. Two cases: (a) without and (b) with ramp metering.

As an example, if $q_{in} + d = 1.2$, q_{cap} (i.e., the total demand exceeds the freeway capacity by 20%) and $q_{con} = 0.95 q_{cap}$ (i.e., the capacity drop due to the congestion is 5%) then $\Delta T_s = 20\%$ results from (5), which demonstrates the importance of ramp metering.

C. Second Answer

We consider (see Fig. 3) two cases of a freeway stretch that includes an on-ramp and an off-ramp, namely (a) without and (b) with metering control. In order to clearly separate the different effects of ramp metering, we will assume here that $q_{con} = q_{cap}$, i.e., no capacity drop due to congestion. Defining the exit rate γ ($0 \leq \gamma \leq 1$) as the portion of the upstream flow that exits at the off-ramp, it is easy to show [2] that the exit flow without control is given by

$$s^{nc} = \frac{\gamma}{1 - \gamma} (q_{cap} - d) \quad (6)$$

while with metering control we have

$$s^{rm} = \gamma \cdot q_{in}. \quad (7)$$

Because $(1 - \gamma)q_{in} + d > q_{cap}$ holds (else the congestion would not have been created), it follows that s^{nc} is less than s^{rm} , hence ramp metering increases the outflow thus decreasing the total

time spent in the system. It is easy to show [2] that the amelioration of the total time spent in this case amounts to

$$\Delta T_s = \gamma \cdot 100. \quad (8)$$

As an example, if the exit rate is $\gamma = 0.05$ then the amelioration is $\Delta T_s = 5\%$. If several upstream off-ramps are blocked by the congestion in absence of ramp metering (which is typically the case in many freeways during rush hours) then the amelioration achievable via introduction of ramp metering is accordingly higher.

Summing up the effects of Sections II-B, II-C in a freeway network, overall amelioration of total time spent by as much as 50% (i.e., halving of the average journey time) may readily result (see Section IV).

D. Further Impacts

The road users choose their respective routes toward their destinations so as to minimize their individual travel times. When a control measure (e.g., ramp metering) is introduced that may change the delay experienced in particular network links (e.g., on-ramps), a portion of the drivers will accordingly change their usual route in order to benefit from, or avoid disbenefits due to the new network conditions. For example, in the case of Fig. 2(b), the upstream flow q_{in} will probably increase while the ramp demand d will decrease as compared to Fig. 2(a). Because the route choice behavior of drivers is predictable to a large extent (Traffic Assignment problem!), ramp metering may also be used so as to impose an operationally desired traffic flow distribution in the overall network, e.g., avoidance of the rat-running phenomenon, increased or decreased utilization of underutilized or overloaded, respectively, parallel arterials, etc. Clearly, the modified routing behavior of drivers should be taken into account in the design and evaluation phases of ramp metering control strategies.

Several field evaluation results (see, e.g., [4]) demonstrate that ramp metering improves the merging behavior of traffic flow at freeway intersections which may have a significant positive impact on traffic safety due to less lane changes and reduced driver stress. Moreover, the increase of network efficiency related to both answers above, is expected to lead to accordingly improved network traffic safety and reduced pollutant emissions to the environment.

III. OVERVIEW OF RAMP METERING STRATEGIES

A. Fixed-Time Strategies

Fixed-time ramp metering strategies are derived off-line for particular times-of-day, based on constant historical demands, without use of real-time measurements. They are based on simple static models. A freeway with several on-ramps and off-ramps is subdivided into sections, each containing one on-ramp. We then have

$$q_j = \sum_{i=1}^j \alpha_{ij} r_i \quad (9)$$

where q_j is the mainline flow of section j , r_i is the on-ramp volume (in veh/h) of section i , and $\alpha_{ij} \in [0, 1]$ expresses the

(known) portion of vehicles that enter the freeway in section i and do not exit the freeway upstream of section j . To avoid congestion

$$q_j \leq q_{cap,j} \quad \forall j \quad (10)$$

must hold, where $q_{cap,j}$ is the capacity of section j . Further constraints are

$$r_{j,min} \leq r_j \leq \min\{r_{j,max}, d_j\} \quad (11)$$

where d_j is the demand while $r_{j,max}$ is the ramp capacity at on-ramp j , and $r_{j,min}$ is the minimum flow that always must be allowed to enter the freeway. This approach was first suggested by Wattleworth [5]. Other similar formulations may be found in [6]–[11]

As an objective criterion, one may wish to maximize the number of served vehicles (which is equivalent to minimizing the total time spent)

$$\sum_j r_j \rightarrow \text{Max} \quad (12a)$$

or to maximize the total traveled distance

$$\sum_j \Delta_j q_j \rightarrow \text{Max} \quad (12b)$$

(where Δ_j is the length of section j), or to balance the ramp queues

$$\sum_j (d_j - r_j)^2 \rightarrow \text{Min}. \quad (12c)$$

These formulations lead to linear-programming or quadratic-programming problems that may be readily solved by use of broadly available computer codes. An extension of these methods that renders the static model (9) dynamic by introduction of constant travel times for each section, was suggested in [12].

The main drawback of fixed-time strategies is that their settings are based on historical rather than real-time data. This may be a crude simplification because of the following.

- Demands are not constant, even within a time-of-day.
- Demands may vary at different days, e.g., due to special events.
- Demands change in the long term leading to “aging” of the optimized settings.
- The portions α_{ij} are also changing in the same ways as demands; in addition, these portions may change due to the drivers’ response to the new optimized signal settings, whereby they try to minimize their individual travel times.
- Incidents and further disturbances may perturb traffic conditions in a nonpredictable way.

Hence, fixed-time ramp metering strategies may lead (due to the absence of real-time measurements) either to overload of the mainstream flow (congestion) or to underutilization of the freeway. In fact, ramp metering is an efficient but also delicate control measure. If ramp metering strategies are not accurate enough, then congestion may not be prevented from forming,

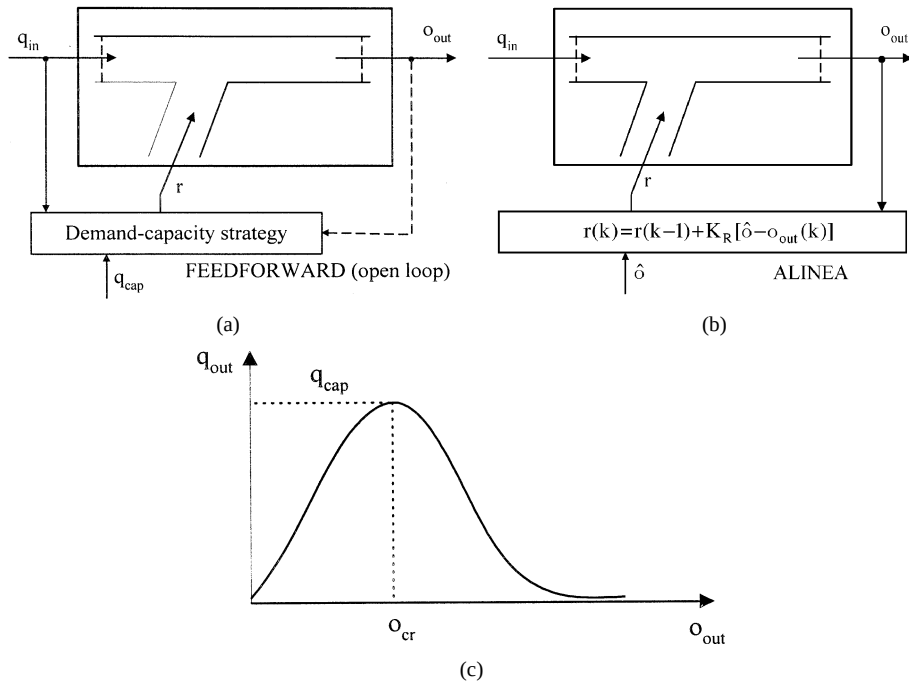


Fig. 4. Local ramp metering strategies. (a) Demand-capacity. (b) ALINEA. (c) Fundamental diagram.

or the mainstream capacity may be underutilized (e.g., due to groundlessly strong metering).

B. Reactive Ramp Metering Strategies

Reactive ramp metering strategies are employed at a tactical level, i.e. in the aim of keeping the freeway traffic conditions close to prespecified set values, based on real-time measurements.

Local Ramp Metering: Local ramp metering strategies make use of traffic measurements in the vicinity of a ramp to calculate suitable ramp metering values. The *demand-capacity strategy* [13], quite popular in North America, reads

$$r(k) = \begin{cases} q_{\text{cap}} - q_{\text{in}}(k-1), & \text{if } o_{\text{out}}(k) \leq o_{\text{cr}} \\ r_{\text{min}}, & \text{else} \end{cases} \quad (13)$$

where (see Fig. 4) q_{cap} is the freeway capacity downstream of the ramp, q_{in} is the freeway flow measurement upstream of the ramp, o_{out} is the freeway occupancy measurement downstream of the ramp, o_{cr} is the critical occupancy (at which the freeway flow becomes maximum), and r_{min} is a prespecified minimum ramp flow value. The strategy (13) attempts to add to the measured upstream flow $q_{\text{in}}(k-1)$ as much ramp flow $r(k)$ as necessary to reach the downstream freeway capacity q_{cap} . If, however, for some reason, the downstream measured occupancy $o_{\text{out}}(k)$ becomes overcritical (i.e., a congestion may form), the ramp flow $r(k)$ is reduced to the minimum flow r_{min} to avoid or to dissolve the congestion. Clearly, (13) does not really represent a closed-loop strategy but an open-loop disturbance-rejection policy [see Fig. 4(a)] which is generally known to be quite sensitive to various nonmeasurable disturbances.

The *occupancy strategy* [13] is based on the same philosophy as the demand-capacity strategy, but it relies on occupancy-based estimation of q_{in} , which may, under certain conditions, reduce the corresponding implementation cost.

An alternative, closed-loop ramp metering strategy (*ALINEA*), suggested in [14], reads

$$r(k) = r(k-1) + K_R [\hat{o} - o_{\text{out}}(k)] \quad (14)$$

where $K_R > 0$ is a regulator parameter and $\hat{o}$ is a set (desired) value for the downstream occupancy [typically, but not necessarily, $\hat{o} = o_{\text{cr}}$ may be set, in which case the downstream freeway flow becomes close to q_{cap} , see Fig. 4(c)]. In field experiments, ALINEA has not been very sensitive to the choice of the regulator parameter K_R .

Note that the demand-capacity strategy reacts to excessive occupancies o_{out} only after a threshold value (o_{cr}) is exceeded, and in a rather crude way, while ALINEA reacts smoothly even to slight differences $\hat{o} - o_{\text{out}}(k)$, and thus it may prevent congestion by stabilizing the traffic flow at a high throughput level. It is easily seen that at a stationary state (i.e., if q_{in} is constant), $o_{\text{out}}(k) = \hat{o}$ results automatically from (14), although no measurements of the inflow q_{in} are explicitly used in the strategy.

The set value $\hat{o}$ may be changed any time, and thus ALINEA may be embedded into a hierarchical control system with set values of the individual ramps being specified in real time by a superior coordination level or by an operator.

All control strategies calculate suitable ramp volumes r . In the case of **traffic-cycle realization** of ramp metering, r is converted to a green-phase duration g by use of

$$g = (r/r_{\text{sat}}) \cdot c \quad (15)$$

where c is the fixed cycle time and r_{sat} is the ramp's saturation flow. The green-phase duration g is constrained by $g \in [g_{\text{min}}, g_{\text{max}}]$, where $g_{\text{min}} > 0$ to avoid ramp closure, and $g_{\text{max}} \leq c$. In the case of an **one-car-per-green realization**, a constant-duration green phase permits exactly one vehicle to pass. Thus, the ramp volume r is controlled by varying the red-phase duration between a minimum (zero) and a maximum

value. Note that ALINEA is also applicable directly to the green or red-phase duration, by combining (14) and (15)

$$g(k) = g(k-1) + K'_R [\hat{o} - o_{\text{out}}(k)] \quad (16)$$

where $K'_R = K_R c / r_{\text{sat}}$. Note also that the values $r(k-1)$ or $g(k-1)$ used on the right-hand side of (14) or (16), respectively, should be the *bounded* values of the previous time step (i.e., after application of the $g_{\min}$ and $g_{\max}$ constraints) in order to avoid the wind-up phenomenon in the regulator.

If the queue of vehicles on the ramp becomes excessive, interference with surface street traffic may occur. This may be detected with suitably placed detectors (on the upstream part of the on-ramp), leading to an override of the regulator decisions to allow more vehicles to enter the freeway and the ramp queue to diminish.

Note that the above specifications and constraints apply in the same way to any ramp metering strategy.

Comparative field trials have been conducted in various countries to assess and compare the efficiency of local ramp metering strategies, see, e.g., [3]. These trials have demonstrated the clear superiority of ALINEA over other local strategies and over the no-control case with regard to any performance criterion: total time spent, total travelled distance, mean speed, mean (daily) congestion duration. Typical local improvements of the total time spent (including the waiting time at the ramps) may reach 20%.

Multivariable Regulator Strategies: Multivariable regulators for ramp metering pursue the same goals as local ramp metering strategies: They attempt to operate the freeway traffic conditions near some pre-specified set (desired) values. While local ramp metering is performed independently for each ramp, based on local measurements, multivariable regulators make use of all available mainstream measurements $o_i(k)$, $i = 1, \dots, n$, on a freeway stretch, to calculate simultaneously the ramp volume values $r_i(k)$, $i = 1, \dots, m$, for all controllable ramps included in the same stretch [15]. This provides potential improvements over local ramp metering because of more comprehensive information provision and because of coordinated control actions. Multivariable regulator approaches to ramp metering have been reported in [15]–[26]. The multivariable regulator strategy *METALINE* may be viewed as a generalization and extension of ALINEA, whereby the metered on-ramp volumes are calculated from (bold variables indicate vectors and matrices)

$$\mathbf{r}(k) = \mathbf{r}(k-1) - \mathbf{K}_1 [\mathbf{o}(k) - \mathbf{o}(k-1)] + \mathbf{K}_2 [\hat{\mathbf{O}} - \mathbf{O}(k)] \quad (17)$$

where $\mathbf{r} = [r_1 \dots r_m]^T$ is the vector of m controllable on-ramp volumes, $\mathbf{o} = [o_1 \dots o_n]^T$ is the vector of n measured occupancies on the freeway stretch, $\mathbf{O} = [O_1 \dots O_m]^T$ is a subset of $\mathbf{o}$ that includes m occupancy locations for which pre-specified set values $\hat{\mathbf{O}} = [\hat{O}_1 \dots \hat{O}_m]^T$ may be given. Note that for control-theoretic reasons, the number of set-valued occupancies cannot be higher than the number of controlled on-ramps. Typically one bottleneck location downstream of each controlled on-ramp is selected for inclusion in the vector $\mathbf{O}$. Finally, $\mathbf{K}_1$ and $\mathbf{K}_2$ are the regulator's constant gain matrices that must be suitably designed, see [15] and [27] for details.

Field trials and simulation results comparing the efficiency of METALINE versus ALINEA lead to the following conclusions [3].

- While ALINEA requires hardly any design effort, METALINE application calls for a rather sophisticated design procedure that is based on advanced control-theoretic methods (LQ optimal control).
- For urban freeways with a high density of on-ramps, METALINE was found to provide no advantages over ALINEA (the later implemented independently at each controllable on-ramp) under recurrent congestion.
- In the case of nonrecurrent congestion (e.g., due to an incident), METALINE performs better than ALINEA due to more comprehensive measurement information.

Some system operators hesitate to apply ramp metering because of the concern that congestion may be conveyed from the freeway to the adjacent street network. In fact, a ramp metering application designed to avoid or reduce congestion on freeways may have both positive and negative effects on the adjacent road network traffic. It is easy to see, based on notions and statements made earlier, that, if an efficient control strategy is applied for ramp metering, the freeway throughput will be generally increased. More precisely, ramp metering at the beginning of the rush hour may lead to on-ramp queues in order to prevent congestion to form on the freeway, which may temporarily lead to diversion toward the urban network. But due to congestion avoidance or reduction, the freeway will be eventually enabled to accommodate a higher throughput, thus attracting drivers from urban paths and leading to an improved overall network performance. This positive impact of ramp metering on both the freeway and the adjacent road network traffic conditions was confirmed in a specially designed field evaluation in the Corridor Périphérique in Paris, France [28].

C. Nonlinear Optimal Ramp Metering Strategies

Prevention or reduction of traffic congestion on freeway networks may dramatically improve the infrastructure efficiency in terms of throughput and total time spent. Congestion on limited-capacity freeways forms, because too many vehicles attempt to use them in a noncoordinated (uncontrolled) way. Once congestion is built up, the outflow from the congestion area is reduced and the off-ramps and interchanges covered by the congestion are blocked, which may in some extreme cases even lead to fatal gridlocks. Reactive ramp metering strategies may be helpful to a certain extent, but, first they need appropriate set values, and, second, their character is more or less local. What is needed for freeway networks or long stretches is a superior coordination level that calculates, in real time, optimal and fair set values from a proactive, strategic point of view. Such an optimal control strategy should explicitly take into account:

- current traffic state both on the freeway and on the on-ramps;
- demand predictions over a sufficiently long time horizon;
- limited storage capacity of the on-ramps;
- ramp metering constraints discussed earlier;

- nonlinear traffic flow dynamics, including the infrastructure's limited capacity;
- any incidents currently present in the freeway network.

Based on this comprehensive information, the control strategy should deliver set values for the overall freeway network over a future time horizon so as to:

- respect all present constraints;
- minimize an objective criterion such as the total time spent in the whole network (including the on-ramps).

Such a comprehensive dynamic optimal control problem may be formulated and solved with moderate computation time by use of suitable solution algorithms (see Section IV).

The nonlinear traffic dynamics may be expressed by use of suitable dynamic models in the form

$$\mathbf{x}(k+1) = \mathbf{f}[\mathbf{x}(k), \mathbf{r}(k), \mathbf{d}(k)] \quad (18)$$

where the state vector $\mathbf{x}$ comprises all traffic densities and mean speeds of 500-m long freeway sections, as well as all ramp queues; the control vector $\mathbf{r}$ comprises all controllable ramp volumes; the disturbance vector $\mathbf{d}$ comprises all on-ramp demands and turning rates (at network bifurcations or at off-ramps). The ramp metering constraints are given by (11) while the queue constraints read

$$l_i(k) \leq l_{i,\max} \quad (19)$$

where l_i are queue lengths (in veh). The total time spent in the whole system over a time horizon K may be expressed

$$T_s = T \sum_{k=0}^K \left[\sum_{i=1}^n \rho_i(k) \cdot \Delta_i + \sum_{i=1}^m l_i(k) \right] \quad (20)$$

where $\rho_i(k)$ is the traffic density (in veh/km) in segment i at time $k \cdot T$.

Thus, for given current (initial) state $\mathbf{x}(0)$ from corresponding measurements or estimates, and given demand predictions $\mathbf{d}(k)$, $k = 0, \dots, K-1$, the problem consists in specifying the ramp flows $\mathbf{r}(k)$, $k = 0, \dots, K-1$, so as to minimize the total time spent (20) subject to the nonlinear traffic flow dynamics (18) and the constraints (11) and (19).

This problem or variations thereof was considered and solved in various works [1], [29]–[36]. Although simulation studies indicate substantial savings of travel time and substantial increase of throughput, advanced control strategies of this kind have not been implemented in the field as yet.

D. Integrated Freeway Network Traffic Control

Modern freeway networks may include different types of control measures. The corresponding control strategies are usually designed and implemented independently, thus failing to exploit the synergistic effects that might result from coordination of the respective control actions. An advanced concept for integrated freeway network control results from suitable extension of the optimal control approach outlined above. More precisely, the dynamic model (18) of freeway traffic flow may be extended to enable the inclusion of further control measures, beyond the ramp metering rates $\mathbf{r}(k)$. Formally $\mathbf{r}(k)$ is then replaced in (18) by a general control input vector $\mathbf{u}(k)$ that comprises all implemented control measures of any type. Such an approach was

implemented in the integrated, generic freeway network control tool AMOC [37] where ramp metering and route guidance are considered simultaneously with promising results, see also [38] and [39].

IV. AN ADVANCED EXAMPLE

A. The Freeway Network Traffic Model

The efficiency and the amelioration potential of nonlinear optimal ramp metering strategies may be demonstrated by means of simulation for a large-scale network with the use of the AMOC generic freeway network control tool. In this case AMOC does not consider routing control measures, but only ramp metering control actions.

The macroscopic model employed for control design purposes is suitable for free flow, critical, and congested traffic conditions. It has two distinct modes of operation. When traffic assignment (routing) aspects of the traffic process are not taken under consideration, it operates in the nondestination oriented mode. When traffic assignment is an issue, it operates in the destination oriented mode. When route guidance measures are not included, then the traffic model does not need to operate in the destination oriented mode, although this is not imperative. Since we are interested here only in ramp metering, the destination oriented mode of operation will not be described (see [40] for details).

The network is represented by a directed graph whereby the links of the graph represent freeway stretches. Each freeway stretch has uniform characteristics, i.e., no on-/off-ramps and no major changes in geometry. The nodes of the graph are placed at locations where a major change in road geometry occurs, as well as at junctions, on-ramps, and off-ramps.

The time and space arguments are discretized. The discrete-time step is denoted by T . A freeway link m is divided into N_m sections of equal length L_m . Each section i of link m at time instant $t = kT$, $k = 0, \dots, K$, is characterized by the following macroscopic quantities: The traffic density $\rho_{m,i}(k)$ (veh/lane/km) is the number of vehicles in section i of link m at time $t = kT$ divided by L_m and by the number of lanes λ_m ; the mean speed $v_{m,i}(k)$ (km/h) is the mean speed of the vehicles included in section i of link m at time kT ; and the traffic volume or flow $q_{m,i}(k)$ (veh/h) is the number of vehicles leaving section i of link m during the time period $[kT, (k+1)T]$, divided by T . The basic equations used for their calculation for each section i of link m at each time step, are

$$\rho_{m,i}(k+1) = \rho_{m,i}(k) + \frac{T}{L_m \cdot \lambda_m} \cdot [q_{m,i-1}(k) - q_{m,i}(k)] \quad (21)$$

$$q_{m,i}(k) = \rho_{m,i}(k) \cdot v_{m,i}(k) \cdot \lambda_m \quad (22)$$

$$\begin{aligned} v_{m,i}(k+1) = v_{m,i}(k) &+ \frac{T}{\tau} \cdot \{V[\rho_{m,i}(k)] - v_{m,i}(k)\} \\ &+ \frac{T}{L_m} \cdot v_{m,i}(k) \cdot [v_{m,i-1}(k) - v_{m,i}(k)] \\ &- \frac{\nu \cdot T \cdot [\rho_{m,i+1}(k) - \rho_{m,i}(k)]}{\tau \cdot L_m \cdot (\rho_{m,i}(k) + \kappa)} \end{aligned} \quad (23)$$

$$V[\rho_{m,i}(k)] = v_{f,m} \cdot \exp \left[-\frac{1}{a_m} \cdot \left(\frac{\rho_{m,i}(k)}{\rho_{cr,m}} \right)^{a_m} \right] \quad (24)$$

where $v_{f,m}$ and $\rho_{cr,m}$ denote the free speed and critical density per lane, respectively, of link m while a_m is a further parameter of the fundamental diagram [see (24)] of link m . Equation (21) is the well-known conservation equation, (22) is the flow equation to be substituted in (1), (23) is an empirical dynamic speed equation with (24) to be replaced therein (for more details see [41]). The mean speeds in the network sections are limited from below by $v_{\min}$, $v_{f,m}$, $\rho_{cr,m}$, a_m , τ , ν , and κ are constant parameters which reflect particular characteristics of a given traffic system and depend upon street geometry, vehicle characteristics, drivers' behavior, etc. For a real-life network these parameters are determined by a validation procedure as the one described in [42].

In order for the speed calculation to take into account the speed decrease caused by merging phenomena, the term $-(\delta T/L_m \lambda_m)[q_\mu(k)v_{m,1}(k)/(\rho_{m,1}(k) + \kappa)]$ is added to the right-hand side of (23), where δ is a further parameter, μ is the merging link and m is the leaving link. In order for the speed reduction due to weaving phenomena, resulting from lane drops in the mainstream, to be considered, the term $-(\phi T/L_m \lambda_m)[\Delta \lambda \rho_{m,N_m}(k)/\rho_{cr,m}]v_{m,N_m}(k)^2$ is also added in (23), where $\Delta \lambda$ is the number of lanes being dropped, and ϕ is a further parameter. For more details on these two additional terms, see [41].

For origin links (e.g., on-ramps), i.e., links that receive traffic demand and subsequently forward it into the freeway network, a simple queue model is used. The outflow $q_o(k)$ of an origin link o is given by

$$q_o(k) = r_o(k) \min \{d_o(k) + l_o(k)/T, q_{\max,o}(k)\} \quad (25)$$

where $d_o(k)$ is the demand flow at time period k at origin o , $l_o(k)$ is the length (in vehicles) of a possibly existing waiting queue at time k , and $r_o(k) \in [r_{\min}, 1]$ is the metering rate for the origin link o at period k , i.e., a control variable. If $r_o(k) = 1$, no ramp metering is applied; if $r_o(k) < 1$, ramp metering becomes active. $q_{\max,o}$ is the maximum outflow at the specific time instant. The latter depends on the density of the mainstream link μ in the following way:

$$q_{\max,o}(k) = \begin{cases} Q_o, & \text{if } \rho_{\mu,1}(k) < \rho_{cr,\mu} \\ Q_o \cdot p(k), & \text{else} \end{cases} \quad (26)$$

where Q_o is the flow capacity of the origin link and $p(k)$ is the portion of Q_o that can enter link μ if $\rho_{1,\mu} > \rho_{cr,\mu}$, where

$$p(k) = \frac{\rho_{\max} - \rho_{\mu,l}(k)}{\rho_{\max} - \rho_{cr,\mu}} \quad (27)$$

with $\rho_{\max}$ the maximum possible density in the network's links. Equations (26), (27) express the reduction of the ramp's outflow capacity caused by mainstream congestion. The conservation equation for an origin link yields

$$l_o(k+1) = l_o(k) + T \cdot [d_o(k) - q_o(k)]. \quad (28)$$

Freeway bifurcations and junctions (including on-ramps and off-ramps) are represented by nodes. Traffic enters a node n through a number of input links and is distributed to a number of output links according to

$$Q_n(k) = \sum_{\mu \in I_n} q_{\mu,N_\mu}(k) \quad (29)$$

$$q_{m,0}(k) = \beta_n^m(k) \cdot Q_n(k), \quad \forall m \in O_n \quad (30)$$

where I_n is the set of links entering node n , O_n is the set of links leaving node n , $Q_n(k)$ is the total traffic volume entering node n at period k , $q_{m,0}(k)$ is the traffic volume that leaves node n via outlink m , and $\beta_n^m(k)$ is the portion of Q_n that leaves the node through link m . $\beta_n^m(k)$ are the turning rates of node n and are assumed known for the entire time horizon. Equations (29) and (30) provide $q_{m,0}(k)$ which is needed in (21) for $i = 1$.

When a node n has more than one leaving links then the upstream influence of density has to be taken into account in the last section of the incoming link [see (23)]. This is provided via

$$\rho_{m,N_{m+1}}(k) = \frac{\sum_{\mu \in O_n} \rho_{\mu,1}^2(k)}{\sum_{\mu \in O_n} \rho_{\mu,1}(k)} \quad (31)$$

where $\rho_{m,N_{m+1}}$ is the virtual downstream density of the entering link m to be used in (23) for $i = N_m$, and $\rho_{\mu,1}(k)$ is the density of the first section of leaving link μ . This quadratic form is used because one congested leaving link may block the entering link even if there is free flow in the other leaving link.

When a node n has more than one entering links, then the downstream influence of speed has to be taken into account according to equation (23). The mean speed value is calculated from

$$v_{m,0}(k) = \frac{\sum_{\mu \in I_n} v_{\mu,N_\mu}(k) \cdot q_{\mu,N_\mu}(k)}{\sum_{\mu \in I_n} q_{\mu,N_\mu}(k)} \quad (32)$$

where $v_{m,0}$ is the virtual speed upstream of the leaving link m that is needed in (23) for $i = 1$.

B. The Constrained Optimal Control Problem

By substituting (22), (29), (30) into (21); (22), (24), (31), (32) into (23); and (25)–(27) into (28), a discrete-time dynamic traffic model in the sense of (18) is formulated for any arbitrary freeway network. The state vector $\mathbf{x}$ consists of all traffic densities and mean speeds of every section i , and of all queues formed at on-ramps. The control vector $\mathbf{r}$ consists of the on-ramp metering rates $r_o(k) \in [r_{o,\min}, 1]$. The disturbance vector $\mathbf{d}$ consists of all on-ramp demands and turning rates at the network bifurcations and off-ramps.

The cost criterion to be minimized is the total time spent T_s [see (20)] in the whole system over a time horizon K , plus two penalty terms, one for consideration of the queue constraints

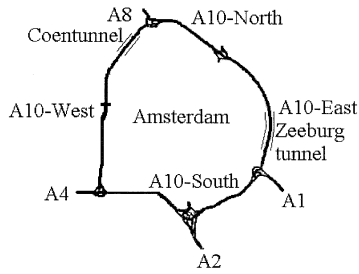


Fig. 5. Amsterdam ringroad.

(19) and the other to suppress high-frequency oscillations of the control variables. More precisely, the cost criterion is given by

$$J = T \cdot \sum_k \left\{ \sum_m \sum_i \rho_{m,i}(k) \cdot L_m \cdot \lambda_m + \sum_o l_o(k) + a_f \sum_o [r_o(k) - r_o(k-1)]^2 + a_w \sum_o \psi[l_o(k)]^2 \right\} \quad (33)$$

with

$$\psi[l_o(k)] = \max\{0, l_o(k) - l_{o,\max}\} \quad (34)$$

where a_w and a_f are appropriately chosen weighting parameters.

The constrained discrete-time optimal control problem described is solved numerically by a powerful feasible-direction optimization algorithm, see [43] for details.

C. Site Description

The previously described approach to network-wide optimal ramp metering has been applied to the Amsterdam ringroad with the use of AMOC.

A sketch of the Amsterdam Orbital Freeway (A10) is shown in Fig. 5. The A10 simultaneously serves local, regional, and inter-regional traffic and acts as a hub for traffic entering and exiting North Holland. There are four main connections with other freeways, the A8 at the North, the A4 at the South-West, the A2 at the South, and the A1 at the South-East. The A10 contains two tunnels, the Coen Tunnel at the North-West of A10 and the Zeeburg Tunnel at the East.

For the purposes of our study we will constrain ourselves to the counterclockwise direction of the A10, which is about 32 km long. There are 21 on-ramps on this freeway, including the connections with the A8, A4, A2, and A1 freeways, and a total of 20 off-ramps, including the junctions with A4, A2, A1, and A8. It is assumed that ramp metering may be performed at each on-ramp, whereby the maximum permissible queue length for the urban on-ramps is set to 20 vehicles, while storage of a maximum of 90 vehicles is permitted on each of the freeway-to-freeway ramps of A8, A4, A2, and A1.

D. No-Control Case

The ring-road was studied for a time horizon of 4 h, from 16:00 until 20:00 using realistic historical demands from the

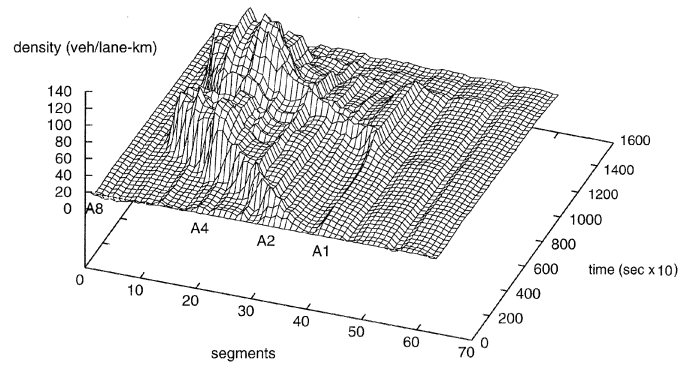


Fig. 6. No control: density.

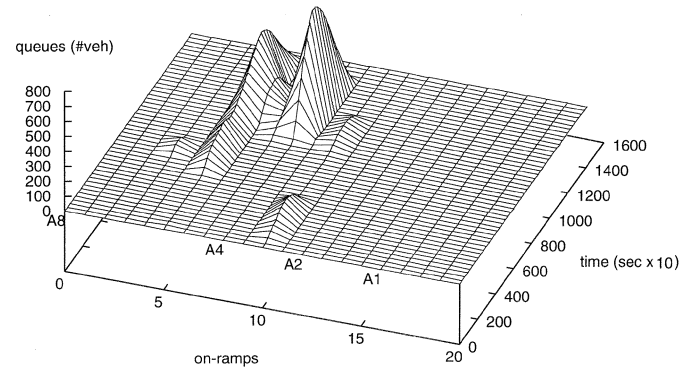


Fig. 7. No control: Ramp queues.

site. This time period includes the evening peak hour. In absence of any control measures, the ring-road is subject to recurrent congestion that is formed downstream of the junctions of A10 with A2 and A1 in A10-South. This congestion propagates backward causing severe congestion to the A10-West. Fig. 6 depicts the density propagation along the freeway sections (section 0 is the first section of A10-West, after the junction of A10 with A8). This congestion causes the formation of large queues at the on-ramps of A10-West as can be seen in Fig. 7, where the queues of each on-ramp are depicted (on-ramp 0 corresponds to A8). Note that on-ramp queues may be created even in absence of ramp metering due to high demand or due to reduced ramp outflow caused by mainstream congestion according to (26), (27). Additionally, the congestion blocks the exits of A10 to A4 and A2 as well as the off-ramps of A10-West and A10-South. As a result, the total time spent over the 4h-horizon is equal to 11 998 veh-h.

E. Optimal-Control Case

When ramp metering is performed at all on-ramps, including the entrances of A8, A4, A2, and A1 to the A10, the congestion is virtually lifted from the network (Fig. 8). The control strategy, aiming at maximizing the time-weighted network outflow (4), succeeds in establishing optimal uncongested traffic conditions on the A10-South and A10-West by applying ramp metering mainly at A1 and A2 at an early stage. In Fig. 9, the queues are mainly occurring at A2 and A1 because these ramps have larger maximum permissible queues (90 vehicles). But

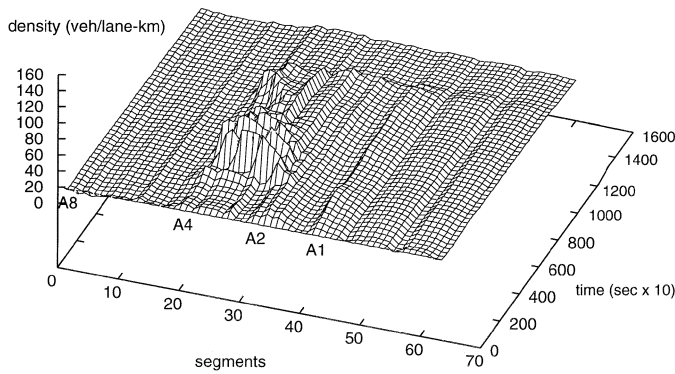


Fig. 8. Optimal control: density.

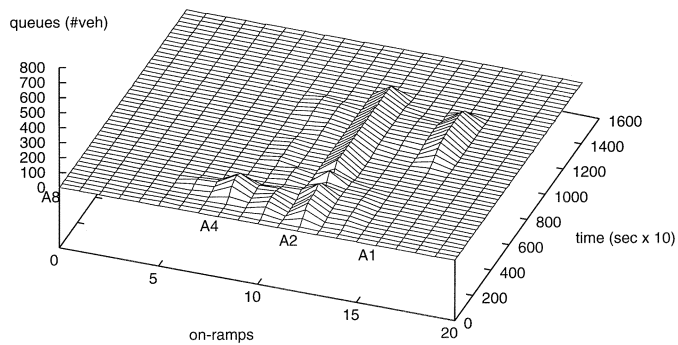


Fig. 9. Optimal control: ramp queues.

since more storage capacity is required for complete elimination of the congestion, further ramp metering is performed at the on-ramps of A10-South and West and at A4, thereby utilising their storage capacity to the fullest extent. The resulting total time spent is 7609 veh·h which is a 36.6% improvement compared to the no-control case. It has to be noted here that this improvement refers to the overall ring-road (not just to the shorter congested stretch) and, also, that it is calculated for the whole time horizon of 4h (not just for the time period where congestion occurs). Therefore there is a systematic underestimation of the amelioration occurring in the critical stretch at the critical time period. Thus, the travel time of drivers involved in the critical stretch during the critical period are reduced even more, without any significant disbenefit to any network user.

A further improvement to the total time spent could be reached with larger maximum permissible queues. Had there been no queue constraints at all, the density profile of Fig. 8 would be completely flat. In fact, the control strategy performs a tradeoff between the queue lengths and the existence of congestion inside the network. Stricter queue constraints result in more degraded traffic conditions inside the freeway due to accordingly reduced control maneuverability.

F. Further Developments

The results of this approach demonstrate its efficiency and feasibility. The computation time is moderate, since for the 4h time horizon the bulk of the 36.6% improvement (more than 32%) was obtained in 20 min on a Sun Ultra5 with a Sparc Ili-360 MHz processor workstation.

Further control measures such as speed control or route guidance may be readily integrated cooperatively due to the flexible nature of the problem formulation.

The control trajectories obtained represent a strategic decision in the sense of providing optimal and fair set values over a long time horizon (e.g., 4 h) for subordinate reactive ramp metering (Section III-B), using e.g. ALINEA. This strategic role can be further enhanced by use of a rolling horizon framework whereby the optimal control problem is solved repeatedly in real time, with updated (measured) initial condition $\mathbf{x}(0)$, updated demand predictions and turning rates $\mathbf{d}(k)$, and with inclusion of possible incidents.

V. CONCLUSIONS

Modern freeway networks' capacity is daily underutilized, particularly during rush hours and at the occurrence of incidents, i.e. when it is most urgently needed, due to the following:

- reduced congestion outflow (see Section II-B);
- reduced off-ramp flow (see Section II-C);
- uncontrolled flow distribution in the overall network (see Section II-D).

The introduction of ramp metering at some particular ramps or particular freeway stretches within the overall network can help to alleviate some local traffic problems and to improve the local traffic conditions. However, the significant amelioration of the global traffic conditions in the overall traffic network calls for **comprehensive** control of **all or most** of the ramps, including the freeway-to-freeway links, in the aim of optimal utilization of the available infrastructure. The limitations of **partial** (rather than comprehensive) ramp metering are as follows.

- 1) The potential benefits of partial ramp metering (according to Figs. 2, 3) may be counterbalanced to some extent by a modified route choice behavior of drivers who attempt the minimization of their individual travel times under the new conditions.
- 2) Individual on-ramps have a limited storage capacity for waiting vehicles; if the on-ramp queue reaches back to the surface street junction, ramp metering control is typically released in order to avoid interference with surface street traffic.
- 3) The freeway network is a common resource for many driver groups with different origins and destinations. Partial ramp metering, by its nature, does not address the strategic problem of optimal utilization of the overall infrastructure, nor does it guarantee a fair and orderly capacity allocation among the ramps.

Comprehensive ramp metering, on the other hand, does not suffer from these shortcomings, first because of complete control of the network traffic flow and its spatial and temporal distribution, and second because of sufficient available storage capacity. In fact, one or a few particular ramps located at a critical bottleneck area may not have sufficient storage capacity to completely avoid the building up of a congestion. However, in case of comprehensive optimal ramp metering in the sense of Section IV, the total available storage space in all ramps and freeway intersections is usually sufficient to effectively and ultimately combat freeway congestion.

It should be emphasized that the implementation and operation cost of a comprehensive ramp metering system is estimated to be rather low as compared to the corresponding infrastructure cost and to the expected benefits in terms of dramatically reduced delays, increased traffic safety, and reduced environmental pollution. It should also be noted that the advanced methodological tools required for efficient operation of such a comprehensive ramp metering system are currently available, see Section IV. The major problem to overcome today, is the inertia of political decision-makers which, on its turn, is mainly due to the lack of understanding of the huge potential of comprehensive ramp metering systems.

We believe that freeway networks will have to be operated as completely controllable systems in the near future, similar to the urban traffic networks, because this is the smartest way to avoid further degradation and fatal gridlocks. The sooner this is realized by the road authorities, the better for the road users who will be the major beneficiaries of this evolution.

ACKNOWLEDGMENT

The authors would like to thank F. Middelham from AVV-Rijkswaterstaat for providing the necessary data for the Amsterdam ringroad.

REFERENCES

- [1] M. Papageorgiou, *Applications of Automatic Control Concepts to Traffic Flow Modeling and Control*. Berlin, Germany: Springer-Verlag, 1983.
- [2] M. Papageorgiou, J. M. Blosseville, and H. Hadj-Salem, "La fluidification des rocade de l'Île de France: Un projet d'importance," *Dynamic Systems and Simulation Lab.*, Tech. Univ. of Crete, Chania, Greece, Internal Rep. 1998-17, 1998.
- [3] M. Papageorgiou, H. Hadj-Salem, and F. Middelham, "ALINEA local ramp metering—summary of field results," in *Transportation Res. Rec.*, 1998, No. 1603, pp. 90–98.
- [4] T. McLean, C. Brader, S. Hangleiter, M. Tsavachidis, C. Damas, B. Maxwell, and P. Barber, Urban integrated traffic control evaluation results, in Deliverable 8.3, Eur. Transport Telematics Project TABASCO, Brussels, Belgium, 1998.
- [5] J. A. Wattleworth, "Peak-period analysis and control of a freeway system," in *Highway Research Rec.*, 1965, No. 157, pp. 1–21.
- [6] L. S. Yuan and J. B. Kreer, "Adjustment of freeway ramp metering rates to balance entrance ramp queues," *Transportation Res.*, vol. 5, pp. 127–133, 1971.
- [7] D. Tabac, "A linear programming model of highway traffic control," in *Proc. 6th Annu. Princeton Conf. on Information Science and Systems*, Princeton, NJ, 1972, pp. 568–570.
- [8] C. F. Wang, "On a ramp-flow assignment problem," *Transportation Sci.*, vol. 6, pp. 114–130, 1972.
- [9] J. J. Wang and A. D. May, "Computer model for optimal freeway on-ramp control," in *Highway Research Rec.*, 1973, No. 469, pp. 16–25.
- [10] I. C. Cheng, J. B. Gruz, and J. G. Paquet, "Entrance ramp control for travel rate maximization in expressways," *Transportation Res.*, vol. 8, pp. 503–508, 1974.
- [11] S. C. Schwartz and H. H. Tan, "Integrated control of freeway entrance ramps by threshold regulation," in *Proc. IEEE Conf. on Decision and Control*, 1977, pp. 984–986.
- [12] M. Papageorgiou, "A new approach to time-of-day control based on a dynamic freeway traffic model," *Transportation Res.*, vol. 14B, pp. 349–360, 1980.
- [13] D. P. Masher, D. W. Ross, P. J. Wong, P. L. Tuan, P. L. Zeidler, and S. Peracek, "Guidelines for design and operating of ramp control systems," SRI, Menid Park, CA, Standford Res. Inst. Rep. NCHRP 3-22, SRI Project 3340, 1975.
- [14] M. Papageorgiou, H. Hadj-Salem, and J.-M. Blosseville, "ALINEA: a local feedback control law for on-ramp metering," in *Transportation Res. Rec.*, 1991, No. 1320, pp. 58–64.
- [15] M. Papageorgiou, J.-M. Blosseville, and H. Hadj-Salem, "Modeling and real-time control of traffic flow on the southern part of Boulevard Périphérique in Paris—Part II: Coordinated on-ramp metering," *Transportation Res.*, vol. 24A, pp. 361–370, 1990.
- [16] L. S. Yuan and J. B. Kreer, "An optimal control algorithm for ramp metering of urban freeways," in *Proc. 6th IEEE Annu. Allerton Conf. on Circuit and System Theory*. Allerton, Illinois, 1968.
- [17] A. Kaya, "Computer and optimization techniques for efficient utilization of urban freeway systems," in *Proc. 5th IFAC World Congress*, Paris, France, 1972, paper 12.1.
- [18] C. H. Knapp, "Traffic estimation and control at bottlenecks," in *Proc. IEEE Int. Conf. on Cybernetics and Society*, Washington, DC, 1972, pp. 469–472.
- [19] L. Isaksen and H. J. Payne, "Suboptimal control of linear systems by augmentation with application to freeway traffic regulation," *IEEE Trans. Automat. Contr.*, vol. 18, pp. 210–219, 1973.
- [20] H. J. Payne, W. S. Meisel, and M. D. Teener, "Ramp control to relieve freeway congestion caused by traffic disturbances," in *Highway Res. Rec.*, 1973, No. 469, pp. 52–64.
- [21] M. Athans, P. K. Houpt, D. Looze, D. Orlhac, S. B. Gershwin, and J. L. Speyer, "Stochastic control of freeway corridor systems," in *Proc. 1975 IEEE Conf. on Decision and Control*, 1975, pp. 676–685.
- [22] D. P. Looze, P. K. Houpt, N. R. Sandell, and M. Athans, "On decentralized estimation and control with application to freeway ramp metering," *IEEE Trans. Automat. Contr.*, vol. 23, pp. 268–275, 1978.
- [23] M. Cremer, "A state feedback approach to freeway traffic control," *Preprints 7th IFAC World Congress*, pp. 1575–1582, 1978.
- [24] N. B. Goldstein and K. S. P. Kumar, "A decentralized control strategy for freeway regulation," *Transportation Res.*, vol. 16B, pp. 279–290, 1982.
- [25] M. Papageorgiou, "Multilayer control system design applied to freeway traffic," *IEEE Trans. Automat. Contr.*, vol. 29, pp. 482–490, 1984.
- [26] L. Benmohamed and S. M. Meerkov, "Feedback control of highway congestion by a fair on-ramp metering," in *Proc. 33rd IEEE Conf. on Decision and Control*, vol. 3, Lake Buena Vista, FL, 1994, pp. 2437–2442.
- [27] C. Diakaki and M. Papageorgiou, "Design and simulation test of integrated corridor control for M8 eastbound corridor in Glasgow," *Dynamic Systems and Simulation Lab.*, Tech. Univ. of Crete, Chania, Greece, Internal Rep. 1995-3, 1995.
- [28] H. Haj-Salem and M. Papageorgiou, "Ramp metering impact on urban corridor traffic: field results," *Transportation Res.*, vol. 29A, pp. 303–319, 1995.
- [29] M. Blinkin, "Problem of optimal control of traffic flow on highways," *Autom. Remote Contr.*, vol. 37, pp. 662–667, 1976.
- [30] M. Papageorgiou and R. Mayr, "Optimal decomposition methods applied to motorway traffic control," *Int. J. Control*, vol. 35, pp. 269–280, 1982.
- [31] N. Bhouri, M. Papageorgiou, and J. M. Blosseville, Optimal control of traffic flow on periurban ringways with application to the Boulevard Périphérique in Paris, in *Preprints 11th IFAC World Congress*, Tallinn, Estonia, vol. 10, pp. 236–243, 1990.
- [32] N. Bhouri, "Commande d'un système de trafic autoroutier: application au Boulevard Périphérique de Paris," Ph.D. dissertation, Univ. Paris-Sud, Centre d'Orsay, France, 1991.
- [33] Y. Stephanedes and K.-K. Chang, "Optimal control of freeway corridors," *ASCE J. Transportation Eng.*, vol. 119, pp. 504–514, 1993.
- [34] H. Zhang, S. Ritchie, and W. Recker, "Some general results on the optimal ramp metering control problem," *Transportation Res.*, vol. 4C, pp. 51–69, 1996.
- [35] A. Kotsialos, M. Papageorgiou, M. Mangeas, and H. Haj-Salem, "Coordinated and integrated control of motorway networks via nonlinear optimal control," *Transportation Res.*, vol. 10C, pp. 65–84, 2002.
- [36] M. Mangeas, H. Haj-Salem, A. Kotsialos, and M. Papageorgiou, "Application of a nonlinear coordinated and integrated control strategy to the Ile-de-France motorway network," unpublished.
- [37] A. Kotsialos, M. Papageorgiou, and A. Messmer, "Optimal coordinated and integrated motorway network traffic control," in *14th Int. Symp. on Transportation and Traffic Theory*, Jerusalem, Israel, 1999, pp. 621–644.
- [38] J. C. Moreno-Banos, M. Papageorgiou, and C. Schäffner, "Integrated optimal flow control in traffic networks," *Eur. J. Op. Res.*, vol. 71, pp. 317–323, 1993.
- [39] B. Ataslar and A. Iftar, "A decentralised control approach for transportation networks," in *Preprints 8th IFAC Symp. on Large Scale Systems*, vol. 2, Patras, Greece, 1998, pp. 348–353.
- [40] A. Messmer and M. Papageorgiou, "METANET: A macroscopic simulation program for motorway networks," *Traffic Eng. Control*, vol. 31, pp. 466–470, 549, 1990.
- [41] M. Papageorgiou, J.-M. Blosseville, and H. Hadj-Salem, "Modeling and real-time control of traffic flow on the southern part of Boulevard Périphérique in Paris—Part I: Modeling," *Transportation Res.*, vol. 24A, pp. 345–359, 1990.

- [42] A. Kotsialos, Y. Pavlis, M. Papageorgiou, and G. Vardaka, Modeling and validation of the Amsterdam Motorway Network, in In DACCORD Deliverable D06.1: Coordinated Control Strategies, Annex B, Eur. Commission, Brussels, Belgium, 1997.
- [43] M. Papageorgiou and M. Marinaki, "A feasible direction algorithm for the numerical solution of optimal control problems," Dynamic Systems and Simulation Lab., Technical Univ. Crete, Chania, Greece, Internal Rep. 1995-4, 1995.



Markos Papageorgiou (M'82–SM'90–F'99) was born in Thessaloniki, Greece, in 1953. He received the Dipl.-Ingenieur and Doktor-Ingenieur (hons) degrees in electrical engineering from the Technical University of Munich, Germany, in 1976 and 1981 respectively.

From 1976 to 1982 he was a Research and Teaching Assistant at the Control Engineering Chair, Technical University of Munich. He was a Free Associate with Dorsch Consult, Munich, Germany, (1982–1988), and with Institute National de Recherche sur les Transports et leur Sécurité (INRETS), Arcueil, France (1986–1988). From 1988 to 1994 he was a Professor of Automation at the Technical University of Munich. Since 1994 he has been a Professor at the Technical University of Crete, Chania, Greece. He was a Visiting Professor at the Politecnico di Milano, Italy, in 1982, at the Ecole Nationale des Ponts et Chaussées, Paris, France from 1985 to 1987, and at MIT, Cambridge, MA in 1997 and 2000; and a Visiting Scholar at the University of Minnesota in 1991 and 1993, University of Southern California in 1993, and the University of California, Berkeley, in 1993, 1997, and 2000. He is the author of the books *Applications of Automatic Control Concepts to Traffic Flow Modeling and Control* (New York: Springer, 1983) and *Optimierung* (Oldenbourg, 1991; 1996), the editor of the *Concise Encyclopedia of Traffic and Transportation Systems* (New York: Pergamon, 1991), and the author or co-author of some 200 technical papers. His research interests include automatic control and optimization theory and applications to traffic and transportation systems, water systems, and further areas.

Dr. Papageorgiou is an Associate Editor of *Transportation Research-Part C* and Chairman of the IFAC Technical Committee on Transportation Systems. He is a member of the Technical Chamber of Greece (TEE). He received the 1983 Eugen-Hartmann award from the Union of German Engineers (VDI), and a Fulbright Lecturing/Research Award (1997).

Apostolos Kotsialos was born in Larissa, Greece, in 1972. He received the Dipl.-Eng. degree in production and management engineering and M.S. degree in operations research from the Technical University of Crete, Greece in 1995 and 1998, respectively.

Since 1995 he has been with the Dynamic Systems and Simulation Laboratory (DSSL) of the Technical University of Crete as a Research Associate where he has been involved in numerous research projects. His main research interests include traffic control of large scale freeway networks, numerical optimization and its applications. He is currently working toward the Ph.D. degree at DSSL working on integrated freeway control.